

INDUSTRIAL ROBOTICS SIMULATION & AUTOMATION

ABB RobotStudio 2023 – Automotive Engine Deburring
and Aeroengine Case Assembly

MSc Robotics – Group Assignment | Cranfield University
Academic Year 2023–2024

TEAM MEMBERS

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Public-facing portfolio report derived from the original academic group assignment.

1. Project Overview

The group assignment involved the design, programming and simulation of two industrial robotic systems using ABB RobotStudio 2023. The first system focused on automotive engine deburring with machine-vision-based quality checking. The second focused on a condensed aeroengine casing assembly process using industrial robots and dedicated end-effectors.

Project objectives

- Select suitable ABB industrial robots for the proposed manufacturing tasks.
- Develop end-effector and workstation concepts using CAD.
- Integrate machine-vision concepts for inspection in the deburring system.
- Create robot workflows and RAPID programs in ABB RobotStudio.
- Simulate the proposed workstations and identify practical implementation issues.

2. Robot Selection

For the automotive deburring workstation, the ABB IRB 2400 was selected as the main industrial robot. For the aeroengine case assembly system, the design used an ABB IRB 6640 together with an ABB IRB 120 for nut-and-bolt assembly tasks.

- ABB IRB 2400 – selected for the automotive engine deburring application.
- ABB IRB 6640 – used for aeroengine casing handling/assembly operations.
- ABB IRB 120 – used for the nut-and-bolt assembly task.

3. Automotive Engine Deburring System

The first workstation was designed around an automotive engine deburring process. The robot was configured to follow programmed paths around the engine component while a machine-vision system was incorporated for quality checking.

Machine vision

- Several COGNEX vision options were compared as part of the system design.
- The COGNEX DS1100 was selected in the report based on factors including scan rate, weight, resolution and integration considerations.

End-effector and CAD design

- A deburring end-effector and camera-mount concept were designed using SolidWorks 2020.
- The CAD design was incorporated into the RobotStudio workstation concept.

4. Aeroengine Case Assembly System

The second workstation represented a condensed aeroengine casing assembly process. The system included a robotic assembly concept, conveyors, a rotating base and dedicated end-effectors for the required operations.

- ABB IRB 6640 used for the main assembly operation.
- ABB IRB 120 used for nut-and-bolt assembly.
- Dedicated end-effector concepts were developed for the assembly tasks.
- A rotating base and conveyor elements were incorporated into the workstation layout.

5. RobotStudio & RAPID Programming

ABB RobotStudio 2023 was used to create the virtual workstations, configure robot paths and simulate the proposed operations. RAPID programming was used to define targets, paths, zones, speeds and instruction sequences.

- Created and configured robot targets and motion paths.
- Used RAPID programming concepts to define the robot workflow.
- Repeated the programmed configuration for the remaining hole/fastener operations.
- Used RobotStudio simulation to review planned robotic movements and workstation behaviour.

6. Results & Practical Considerations

The simulated workstations demonstrated the proposed robotic workflows and provided a way to identify practical issues before physical implementation.

- Path accuracy and robot programming requirements.
- Potential singularity and tool-centre-point considerations.
- End-effector weight and interference considerations.
- Workstation and component interference.
- Challenges associated with achieving realistic simulation behaviour.

The academic report notes that, because of program constraints, 46 nuts and bolts were successfully placed rather than all 48 planned positions.

7. My Contribution – Vrushab Sawant

My contribution to the group assignment focused on robot selection, potential implementation issues, system workflow and support for programming the Part B assembly system in ABB RobotStudio.

- Researched and documented the selection of suitable industrial robots for the proposed applications.
- Worked on robot selection considerations for the automotive deburring and aeroengine assembly systems.
- Helped with programming the Part B system in ABB RobotStudio.
- Contributed to the overall system workflow and process description.
- Researched and documented potential implementation issues and practical challenges.

8. Tools & Technologies

ABB RobotStudio 2023 • RAPID • SolidWorks 2020 • Industrial Robotics • Robot Workcell Simulation • Machine Vision Concepts • CAD / End-Effector Design

9. Portfolio Note

This document is a condensed, public-facing portfolio version of an MSc Robotics group assignment. It presents the project scope and my individual contribution without reproducing the full academic report. The original work was completed as a group project; the contribution section distinguishes the work specifically attributed to Vrushab Sawant in the original report.

SELECTED PROJECT VISUALS

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- b. Palletizing : The IRB 6640 can be set up to palletize products efficiently and assist in packaging automation, regardless of whether they are finished goods or components.
- c. Assembly: Accurate and efficient product assembly is ensured by using the IRB 6640.

Table 1.3 Robot Model Specifications IRB 6640 [2]

Robot Version	Reach (mm)	handling Capacity(kg)	Center of Gravity(kg)	Wrist torque(Nm)
IRB 6640-2.35/2.55	2.55	2.35	300	1324
IRB 6640-1.85/2.8	2.80	1.85	300	1206
Number of axes	6			
Protection	IP67-Foundry Prime 2			
Mounting	Floor			
Controller	IRC5 Single Cabinet			

Table 1.4 Range of robot IRB 6640 [2]

Axis Movement	Working Range	Axis max. speed
Axis 1 rotation	-170° to +170°	100 to -110°/s
Axis 2 arm	+85° to -65°	90°/s
Axis 3 arm	+70° to -180°	90°/s
Axis 4 rotation	-300° to +300°	170 to -190°/s
Axis 5 bend	+120° to -120°	120 to -140°/s
Axis 6 turn	+360° to -360°	190 to -235°/s

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such that the camera stand and end effector may be mounted to a single robot and function as a single unit.

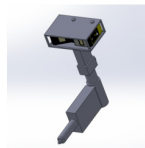


Fig 3.2 Assembly of End-effector with machine vision camera mount



Fig 3.3 COGNEX-DS1100 camera CAD model

3.2. End-effector of aeroengine assembly

The dimensions of upper and lower case were the initial factor that was taken into account while designing the end-effector for Task 2. After importing the upper and lower cases into SolidWorks 2020, measurements were gathered and an off-the-shelf gripper was created. In order to facilitate picking and placing of cases with ease, the

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4. Overall system design

For task 1: The ABB IRB 2400 was selected and the workstation was created. The robot's location is fixed at the real import point, and its controller is positioned at the robot's rear. Through the geometry, the engine stand and block were imported and placed parallel to the robot's front perspective. To guarantee that the robot's arm length reaches the engine block on all sides while deburring, the robot is positioned parallel to the engine block and stand. This also helps to shorten the cycle time. The end-effector CAD model SAT file was imported to the ABB Studio 2023 (fig 4.1).

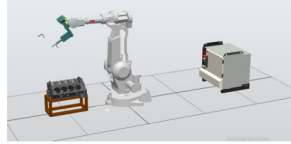


Fig 4.1 Workstation of Deburring system of automotive engine block

For task 2: ABB IRB 6640 was selected and imported along with its controller. The euro plates that would hold the upper and lower cases were imported from the library legacy mechanism models. After being imported, the lower and upper cases were positioned on the left and right of the euro plate, respectively. On the workstation, the rotating base was imported and then made into a rotating mechanism at the global origin. The grabber's SAT file which will be used for picking and placing of cases was imported, and it was positioned on the robot. Conveyor belts, IRB 120, and a sorter were

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For task 2 : Upon importing the CAD models for each end-effector, a workstation was built. For the purpose of picking and placing the lower and upper cases on the rotating base, the grabber was developed into a smart component, and targets and paths were built (fig 5.3). Following the case placement on the rotating base, the workstation conveyor, sorter, and IRB120 were adjusted appropriately. We scaled up the conveyors to 1.3 factor based on the IRB 120 model. Nut and Bolts were placed on the sorter machine. Two smart components were added which will help nut and bolt move onto their conveyors respectively. Each conveyor has a plane sensor at its tip that helps the bolt and nut stop so the robot can pick them up. The end effectors for nut and bolt were positioned on the IRB 120 and converted into smart components, their targets and paths were created so that it can pick and place the nut and bolt (fig 5.4). With the help of FOR loop we repeated the configuration for the remaining 47 holes.



Fig 5.3 Pick and Place of Lower and Upper Case

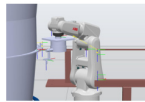


Fig 5.4 Pick and Place of Bolt Assembly

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